

第十五章 正态分布专项选择题(20 道)

(Special Multiple-Choice Questions on Normal Distribution (20 Questions))

一、基础概念类(Basic Concepts)

1. 下列关于连续型随机变量的说法, 正确的是() (Which of the following statements about continuous random variables is correct?)

- A. 取值为有限个离散点(Takes finite discrete values)
- B. 单个取值的概率为 0(Probability of a single value is 0)
- C. 包含离散型随机变量(Includes discrete random variables)
- D. 取值只能是全体实数(Can only take values of the entire real axis)

答案:B(Answer: B)

解析: 连续型随机变量取值充满区间, 单个取值概率为 0; A、C 是离散型特征, D 错误(如身高仅为正区间)(Explanation: Continuous random variables take values filling an interval, and the probability of a single value is 0; A and C are characteristics of discrete types, D is incorrect (e.g., height is only a positive interval)).

2. 若 $X \sim N(\mu, \sigma^2)$, 正态曲线的对称轴是() (If $X \sim N(\mu, \sigma^2)$, the axis of symmetry of the normal curve is)

- A. $x = 0$
- B. $x = \mu$
- C. $x = \sigma$
- D. $x = \mu + \sigma$

答案:B(Answer: B)

解析: 正态曲线关于 $x = \mu$ 对称, 仅标准正态分布($\mu = 0$)关于 $x = 0$ 对称(Explanation: The normal curve is symmetric about $x = \mu$, only the standard normal distribution ($\mu = 0$) is symmetric about $x = 0$).

3. 标准正态分布的参数是() (The parameters of the standard normal distribution are)

- A. $\mu = 0, \sigma = 0$
- B. $\mu = 1, \sigma = 1$
- C. $\mu = 0, \sigma = 1$
- D. $\mu = 1, \sigma = 0$

答案:C(Answer: C)

解析: 标准正态分布定义为 $\mu = 0, \sigma = 1$, $\sigma > 0$ (标准差不能为 0)(Explanation: The standard normal distribution is defined as $\mu = 0, \sigma = 1$, and $\sigma > 0$ (standard deviation cannot be 0)).

4. 正态分布 $N(\mu, \sigma^2)$ 中, σ 的取值范围是() (In the normal distribution $N(\mu, \sigma^2)$, the range of σ is)

- A. $\sigma < 0$
- B. $\sigma = 0$
- C. $\sigma \geq 0$
- D. $\sigma > 0$

答案:D(Answer: D)

解析: σ 是标准差, 必须大于 0(若 $\sigma = 0$, 分布退化为单点 μ , 无意义)(Explanation: σ is the standard deviation and must be greater than 0 (if $\sigma = 0$, the distribution degenerates to a single point μ , which is meaningless)).

二、参数意义类(Parameter Meanings)

5. 若 $X \sim N(\mu_1, \sigma^2)$, $Y \sim N(\mu_2, \sigma^2)$ ($\mu_1 \neq \mu_2$), 则 X 与 Y 的正态曲线() (If $X \sim N(\mu_1, \sigma^2)$ and $Y \sim N(\mu_2, \sigma^2)$ ($\mu_1 \neq \mu_2$), then the normal curves of X and Y)

- A. 位置相同, 肥瘦不同(Same position, different fatness/leanness)
- B. 位置不同, 肥瘦相同(Different positions, same fatness/leanness)
- C. 位置相同, 肥瘦相同(Same position, same fatness/leanness)
- D. 位置不同, 肥瘦不同(Different positions, different fatness/leanness)

答案:B(Answer: B)

解析: μ 决定位置, σ 决定肥瘦; $\mu_1 \neq \mu_2$ 则位置不同, σ 相同则肥瘦相同(Explanation: μ determines the position, σ determines the fatness/leanness; $\mu_1 \neq \mu_2$ means different positions, and the same σ means the same fatness/leanness)。

6. 若 $X \sim N(\mu, \sigma_1^2)$, $Y \sim N(\mu, \sigma_2^2)$ ($\sigma_1 < \sigma_2$), 则 X 的正态曲线比 Y 的() (If $X \sim N(\mu, \sigma_1^2)$ and $Y \sim N(\mu, \sigma_2^2)$ ($\sigma_1 < \sigma_2$), then the normal curve of X is compared to that of Y)

- A. 更矮胖(Shorter and fatter)
- B. 更高瘦(Taller and leaner)
- C. 位置更左(More to the left)
- D. 位置更右(More to the right)

答案:B(Answer: B)

解析: σ 越小, 曲线越“瘦高”; $\sigma_1 < \sigma_2$, 故 X 的曲线更高瘦(Explanation: The smaller σ is, the “leaner and taller” the curve; $\sigma_1 < \sigma_2$, so the curve of X is taller and leaner)。

7. 已知 $X \sim N(5, 9)$, 则 X 的均值和方差分别是() (Given $X \sim N(5, 9)$, the mean and variance of X are respectively)

- A. 5, 3
- B. 5, 9
- C. 3, 5
- D. 9, 5

答案:B(Answer: B)

解析: $N(\mu, \sigma^2)$ 中, $\mu = 5$ 是均值, $\sigma^2 = 9$ 是方差(Explanation: In $N(\mu, \sigma^2)$, $\mu = 5$ is the mean and $\sigma^2 = 9$ is the variance)。

8. 关于正态分布的参数 μ 和 σ , 下列说法错误的是() (Which of the following statements about the parameters μ and σ of the normal distribution is incorrect?)

- A. μ 决定曲线的中心位置(μ determines the central position of the curve)
- B. σ 决定曲线的肥瘦程度(σ determines the fatness/leanness of the curve)
- C. μ 可以是任意实数(μ can be any real number)
- D. σ 越大, 数据越集中(The larger σ is, the more concentrated the data is)

答案:D(Answer: D)

解析: σ 越大, 数据越分散(曲线越矮胖)(Explanation: The larger σ is, the more scattered the data is (the curve is shorter and fatter))。

三、3σ原则与概率计算类(3σ Principle and Probability Calculation)

9. 若 $X \sim N(\mu, \sigma^2)$, 则 $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx ()$ (If $X \sim N(\mu, \sigma^2)$, then $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx)$

- A. 68.27%
- B. 95.45%
- C. 99.73%
- D. 100%

答案:B(Answer: B)

解析: 3σ原则中, $\mu \pm 2\sigma$ 对应的概率为 95.45%(Explanation: In the 3σ principle, the probability corresponding to $\mu \pm 2\sigma$ is 95.45%)。

10. 已知 $X \sim N(0, 1)$ (标准正态分布), 则 $P(|X| \leq 1) \approx ()$ (Given $X \sim N(0, 1)$ (standard normal distribution), then $P(|X| \leq 1) \approx)$

- A. 68.27%
- B. 95.45%
- C. 99.73%
- D. 50%

答案:A(Answer: A)

解析: $|X| \leq 1$ 即 $-1 \leq X \leq 1$, 标准正态分布下该区间概率为 68.27%(Explanation: $|X| \leq 1$ means $-1 \leq X \leq 1$, and the probability of this interval under the standard normal distribution is 68.27%)。

11. 已知 $X \sim N(10, 4)$, 则 $P(X > 14) \approx ()$ (Given $X \sim N(10, 4)$, then $P(X > 14) \approx)$

- A. 2.28%
- B. 15.87%
- C. 0.14%
- D. 50%

答案:A(Answer: A)

解析: $14 = 10 + 2\sigma$, $P(X > 14) = \frac{1-95.45\%}{2} = 2.275\% \approx 2.28\%$ (Explanation: $14 = 10 + 2\sigma$, $P(X >$

$14) = \frac{1-95.45\%}{2} = 2.275\% \approx 2.28\%$)。

12. 若 $X \sim N(\mu, \sigma^2)$, 则 $P(X < \mu - a)$ 与 $P(X > \mu + a)$ 的关系是() (If $X \sim N(\mu, \sigma^2)$, the relationship between $P(X < \mu - a)$ and $P(X > \mu + a)$ is)

- A. 相等(Equal)
- B. 前者大(The former is larger)
- C. 后者大(The latter is larger)
- D. 无法确定(Cannot be determined)

答案:A(Answer: A)

解析: 正态曲线关于 $x = \mu$ 对称, 故 $P(X < \mu - a) = P(X > \mu + a)$ (Explanation: The normal curve is symmetric about $x = \mu$, so $P(X < \mu - a) = P(X > \mu + a)$)。

13. 已知测量误差 $X \sim N(0, 9)$, 则误差落在 (3, 6) 内的概率约为() (Given the measurement error $X \sim N(0, 9)$, the probability that the error falls within (3, 6) is approximately)

- A. 13.59%
- B. 2.28%

C. 68.27%

D. 95.45%

答案:A(Answer: A)

解析: $(3,6) = (\mu + \sigma, \mu + 2\sigma)$, 概率为 $\frac{95.45\% - 68.27\%}{2} = 13.59\%$ (Explanation: $(3,6) = (\mu + \sigma, \mu + 2\sigma)$,

and the probability is $\frac{95.45\% - 68.27\%}{2} = 13.59\%$)。

四、应用与性质类(Applications and Properties)

14. 下列随机变量中, 最可能服从正态分布的是()(Which of the following random variables is most likely to follow a normal distribution?)

A. 掷骰子 1 次的点数(Number of points in one roll of a die)

B. 投篮 10 次的命中次数(Number of successful shots in 10 attempts)

C. 同年龄儿童的身高(Height of children of the same age)

D. 学生的学号(Student ID numbers)

答案:C(Answer: C)

解析: A、B 是离散型随机变量; D 是人工编号, 无随机性; C 是连续型且中间集中, 符合正态分布 (Explanation: A and B are discrete random variables; D is an artificial number with no randomness; C is continuous and concentrated in the middle, conforming to the normal distribution)。

15. 正态曲线与 x 轴围成的面积为()(The area under the normal curve and above the x-axis is)

A. 0

B. 0.5

C. 1

D. 不确定(Uncertain)

答案:C(Answer: C)

解析: 概率分布的基本性质: 总概率为 1, 对应曲线与 x 轴围成的面积为 1 (Explanation: Basic property of probability distribution: total probability is 1, corresponding to the area under the curve and above the x-axis being 1)。

16. 下列关于正态分布的说法, 正确的是()(Which of the following statements about the normal distribution is correct?)

A. 连续型随机变量的取值是有限个(Continuous random variables take finite values)

B. 连续型随机变量单个取值的概率大于 0 (The probability of a single value of a continuous random variable is greater than 0)

C. σ 越小, 数据越分散(The smaller σ is, the more scattered the data is)

D. 正态曲线关于 $x = \mu$ 对称(The normal curve is symmetric about $x = \mu$)

答案:D(Answer: D)

解析: A 错误(连续型取值充满区间); B 错误(单个取值概率为 0); C 错误(σ 越小数据越集中) (Explanation: A is incorrect (continuous values fill an interval); B is incorrect (probability of a single value is 0); C is incorrect (the smaller σ is, the more concentrated the data is))。

17. 某产品尺寸 $X \sim N(\mu, \sigma^2)$, 按 3σ 原则, 尺寸超出 $\mu \pm 3\sigma$ 的概率约为()(For a product's dimension $X \sim N(\mu, \sigma^2)$, according to the 3σ principle, the probability that the dimension exceeds $\mu \pm 3\sigma$ is approximately)

A. 0.27%

- B. 2.28%
- C. 15.87%
- D. 68.27%

答案:A(Answer: A)

解析: 3σ 原则中, 超出 $\mu \pm 3\sigma$ 的概率为 $1 - 99.73\% = 0.27\%$ (Explanation: In the 3σ principle, the probability of exceeding $\mu \pm 3\sigma$ is $1 - 99.73\% = 0.27\%$)。

18. 已知 $X \sim N(2, 1)$, 且 $P(2 < X \leq 2.5) = 0.36$, 则 $P(X > 2.5) = ()$ (Given $X \sim N(2, 1)$ and $P(2 < X \leq 2.5) = 0.36$, then $P(X > 2.5) = ()$)

- A. 0.14
- B. 0.36
- C. 0.5
- D. 0.86

答案:A(Answer: A)

解析: $P(X > 2) = 0.5$, 故 $P(X > 2.5) = 0.5 - 0.36 = 0.14$ (Explanation: $P(X > 2) = 0.5$, so $P(X > 2.5) = 0.5 - 0.36 = 0.14$)。

19. 若 $X \sim N(10, 4)$, $Y \sim N(10, 9)$, 则 $P(8 < X < 12)$ 与 $P(8 < Y < 12)$ 的大小关系是() (If $X \sim N(10, 4)$ and $Y \sim N(10, 9)$, the size relationship between $P(8 < X < 12)$ and $P(8 < Y < 12)$ is)

- A. 前者大(The former is larger)
- B. 后者大(The latter is larger)
- C. 相等(Equal)
- D. 无法确定(Cannot be determined)

答案:A(Answer: A)

解析: X 的 $\sigma = 2$, Y 的 $\sigma = 3$; X 的曲线更瘦高, $[8, 12] = [\mu - \sigma, \mu + \sigma]$ 区间的概率更大 (Explanation: $\sigma = 2$ for X and $\sigma = 3$ for Y ; the curve of X is leaner and taller, and the probability of the interval $[8, 12] = [\mu - \sigma, \mu + \sigma]$ is larger)。

20. 关于 3σ 原则, 下列说法错误的是() (Which of the following statements about the 3σ principle is incorrect?)

- A. 是正态分布的重要性质(It is an important property of the normal distribution)
- B. 数据几乎都落在 $\mu \pm 3\sigma$ 内(Almost all data falls within $\mu \pm 3\sigma$)
- C. 是“绝对界限”, 数据一定不超出 $\mu \pm 3\sigma$ (It is an “absolute boundary”, and data must not exceed $\mu \pm 3\sigma$)
- D. 常用于质量控制(It is often used in quality control)

答案:C(Answer: C)

解析: 3σ 原则是“近似概率”, 正态变量仍可能超出 $\mu \pm 3\sigma$, 只是概率极低(Explanation: The 3σ principle is an “approximate probability”, and normal variables may still exceed $\mu \pm 3\sigma$, but the probability is extremely low)。